



Modelling Financial Markets

Chapter 14: Deep Learning and Time-Series Foundation Models



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ ACF: AutoCorrelation Function
- ▣ ACI: Adaptive Conformal Inference
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARFIMA: AutoRegressive Fractionally Integrated Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ CI: Confidence Interval
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CPU: Central Processing Unit
- ▣ CRPS: Continuous Ranked Probability Score
- ▣ CRSP: Center for Research in Security Prices
- ▣ DM: Diebold–Mariano (test)
- ▣ DQ: Dynamic Quantile (test)
- ▣ ES: Expected Shortfall
- ▣ ETF: Exchange-Traded Fund
- ▣ FHS: Filtered Historical Simulation
- ▣ FRED: Federal Reserve Economic Data
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GIFT: General Time Series Forecasting Model Evaluation (benchmark GIFT-Eval)
- ▣ GPD: Generalised Pareto Distribution
- ▣ GRU: Gated Recurrent Unit
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HS: Historical Simulation
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator

Acronyms used in this material (2/2)

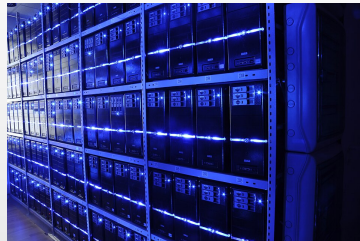
- ▣ LLM: Large Language Model
- ▣ LR: Likelihood Ratio (test)
- ▣ LSTM: Long Short-Term Memory (network)
- ▣ MCS: Model Confidence Set
- ▣ MDE: Minimum Detectable Effect
- ▣ MLP: MultiLayer Perceptron
- ▣ MSE: Mean Squared Error
- ▣ NBER: National Bureau of Economic Research
- ▣ PIT: Probability Integral Transform
- ▣ QLIKE: Quasi-LIKElihood (loss function)
- ▣ QS: Quantile Score
- ▣ RFF: Random Fourier Features
- ▣ RNN: Recurrent Neural Network
- ▣ RV: Realised Variance
- ▣ SPA: Superior Predictive Ability (test)
- ▣ TS: Time Series (in the name TS-Arena)
- ▣ US: United States
- ▣ VAR: Vector AutoRegression

Chapter Route and Outcomes

- Question: do deep networks and time-series **foundation models** forecast **returns**, **volatility** and **tail risk** better than the classical models of Chapters 5–9?
 - ▶ **Deep learning**: models built from many layers of simple non-linear units, trained on data; since 2024, foundation models pretrained on billions of observations
 - ▶ Answered with the evaluation tools of Chapter 8: strong benchmarks, loss differentials, backtests
- Route of the chapter
 - ▶ Part I: networks for sequences, attention, foundation models, Test 1 (returns)
 - ▶ Part II: Test 2 (realised volatility), Test 3 (VaR 1% and ES 2.5%), case study, AI for scientific discovery
- By the end you can
 - ▶ explain vanishing gradients as VAR impulse responses and the LSTM cell state as an AR(1)
 - ▶ turn a zero-shot quantile grid into point, volatility and VaR/ES forecasts
 - ▶ test a new model correctly: nesting, multiplicity, calibration, contamination and power

Three Forecasting Targets

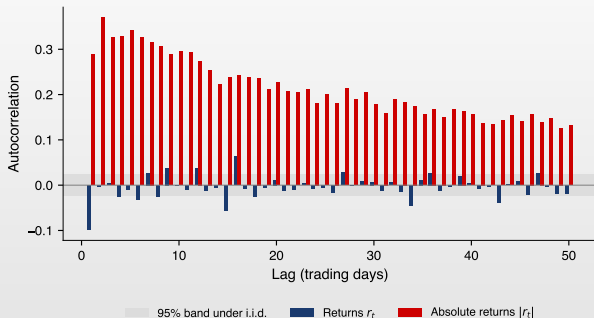
- ▣ **Returns** r_t (conditional mean)
 - ▶ Very low signal-to-noise ratio; predictable patterns are traded away (Chapter 2)
 - ▶ Benchmark: the zero forecast
- ▣ **Volatility** (conditional variance)
 - ▶ Persistent and clustered (Chapters 1 and 5): there is a signal to learn
 - ▶ Benchmark: log-HAR
- ▣ **Tail-risk measures**: VaR 1% and ES 2.5%
 - ▶ Need the shape of the lower tail, not just the centre of the distribution (Chapter 7)
 - ▶ Benchmark: FHS



A data centre: deep networks are trained on racks of such servers

Photo: [BalticServers.com](#) (2013); CC BY-SA 3.0; [Wikimedia Commons](#)

Where Is the Signal?



- S&P 500 daily returns since 2000 (6 718 days), lags 1–50; grey band: $\pm 1.96/\sqrt{T} = \pm 0.024$ under independence
- Returns: lag-1 autocorrelation -0.099; 15 of 50 lags outside the band, small and of changing sign: little to learn about the mean
- Absolute returns: 0.29 at lag 1, still 0.13 at lag 50; all 50 lags outside the band: persistent volatility clustering
- Prediction for the chapter: flexible models can find the volatility signal, not much in the mean

From the Multilayer Perceptron to Deep Learning

- MLP (multilayer perceptron, Chapter 13): $h^{(\ell)} = \varphi \left(W^{(\ell)} h^{(\ell-1)} + b^{(\ell)} \right)$,
 $h^{(0)} = x$; φ : ReLU, tanh or sigmoid $\sigma(z) = 1/(1 + e^{-z})$
 - ▶ Backpropagation (Rumelhart, Hinton & Williams (1986)) and Adam (Kingma & Ba (2015)); loss MSE for points, pinball $(u - 1\{y < q\})(y - q)$ for a quantile of level u
- Universal approximation (Hornik, Stinchcombe & White (1989)): silent on the sample size needed, the binding constraint with one financial series
 - ▶ Regularisation: early stopping, weight decay, ensembles over seeds; the validation block must follow the training block
- 2024 Nobel Prize in Physics to Hopfield and Hinton (LeCun, Bengio & Hinton (2015) for the methods); in finance, Gu, Kelly & Xiu (2020) gain on a cross-section of thousands of stocks, not on one index



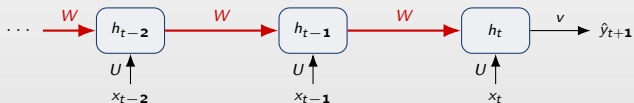
Geoffrey Hinton, Nobel laureate 2024

Photo: Arthur Petron (2024); CC BY-SA 4.0;

Wikimedia Commons

RNN and LSTM in One Slide

- ▣ **Recurrent neural network (RNN):** $h_t = \tanh(W h_{t-1} + U x_t + b)$, $\hat{y}_{t+1} = v^\top h_t + c$
 - ▶ The same weights at every step: unrolled over time, a very deep network with shared weights
- ▣ **Long short-term memory (LSTM, Hochreiter & Schmidhuber (1997)):** gates $f_t, i_t, o_t = \sigma(W_\bullet x_t + U_\bullet h_{t-1} + b_\bullet)$, candidate $g_t = \tanh(W_g x_t + U_g h_{t-1} + b_g)$
 - ▶ Cell state $c_t = f_t \odot c_{t-1} + i_t \odot g_t$, output $h_t = o_t \odot \tanh(c_t)$; $\odot$: element-by-element product
 - ▶ f_t : forget gate, i_t : input gate, o_t : output gate; GRU (gated recurrent unit): a lighter variant with two gates



Vanishing Gradients: a Bound and a VAR(1)

- Backpropagation through time multiplies one Jacobian per step

▶ $\frac{\partial h_T}{\partial h_{T-k}} = \prod_{j=0}^{k-1} D_{T-j} W, \quad D_t = \text{diag}(\varphi'(Wh_{t-1} + Ux_t + b))$

- ▶ Submultiplicativity: $\|\partial h_T / \partial h_{T-k}\|_2 \leq (\gamma_\varphi \|W\|_2)^k$, $\gamma_\varphi = \sup_z |\varphi'(z)|$ ($= 1$ for $\tanh$): $\gamma_\varphi \|W\|_2 < 1$ suffices for vanishing (Pascanu, Mikolov & Bengio (2013))

- Linearise at a fixed point h^* : $h_t - h^* \approx A(h_{t-1} - h^*) + DUx_t$, $A = DW$: a VAR(1) (vector autoregression) in the hidden state

- ▶ $\partial h_T / \partial h_{T-k} = A^k$ is its impulse response at horizon k : stable VAR, $\rho(A) < 1$, gradients decay like $\rho(A)^k$; $\rho(A) > 1$, they explode

- ▶ A stable RNN near its fixed point is a short-memory model, like any stationary VAR (Seminar A1; details in the

▶ Appendix)

- Scalar case $|w|^k$: $0.9^{20} = 0.12$, $0.9^{50} = 0.005$ (Bengio, Simard & Frasconi (1994))

- ▶ Remedies: clipping the gradient norm (explosion); gating, as in the LSTM (vanishing)

Why the LSTM Remembers

- Along the cell state the only multiplier is the forget gate
 - ▶ $\frac{\partial c_T}{\partial c_{T-k}} = \prod_{j=0}^{k-1} f_{T-j}$ (direct path); $c_t = f_t \odot c_{t-1} + i_t \odot g_t$: a random-coefficient AR(1)
- Constant forget gate $f = \sigma(b_f)$: an AR(1) with coefficient f
 - ▶ $b_f = 0$: $f = 0.5$, $0.5^{50} \approx 9 \times 10^{-16}$
 - ▶ $b_f = 3$: $f = 0.953$, $0.953^{50} \approx 0.09$; half-life 14 steps
- Short memory by construction: $\text{Corr}(c_t, c_{t-k}) = f^k$ decays geometrically; long memory decays hyperbolically, $\rho_k \sim Ck^{2d-1}$
 - ▶ Fitted to the S&P 500 absolute returns, $f = 0.984$ predicts 0.027 at lag 150 against 0.059 observed (▶ [Appendix](#))

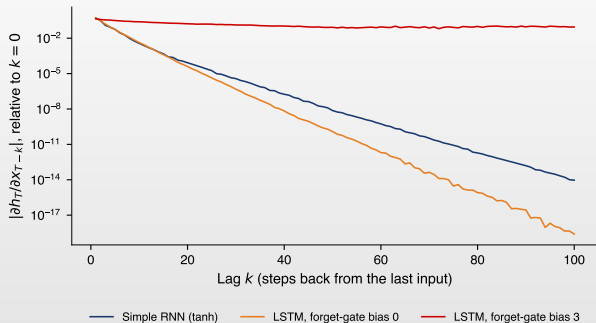


Sepp Hochreiter, co-inventor of the LSTM

Photo: Eulenreich (2015); CC BY-SA 4.0;

[Wikimedia Commons](#)

The Vanishing Gradient, Measured

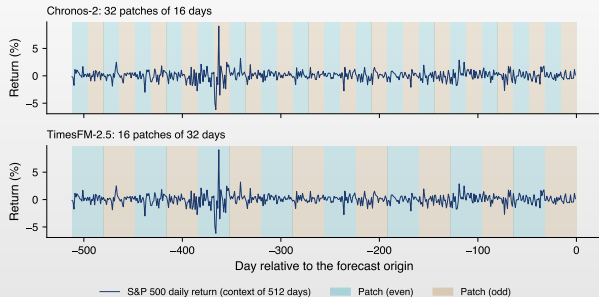


- $|\partial h_T / \partial x_{T-k}|$ by automatic differentiation; 16 hidden units, 50 random initialisations, same Normal inputs
- Simple RNN: 8.3×10^{-5} at lag 20, 7.4×10^{-9} at lag 50: nothing is learned from inputs more than a few weeks back
- LSTM, forget-gate bias 0: 1.2×10^{-10} at lag 50, no better than the RNN ($f \approx 0.5$ halves the memory at every step)
- LSTM, bias 3: 8.4% at lag 50, as $0.953^{50} \approx 0.09$ predicts; initialise the forget-gate bias at a positive value

Attention, Positions and Patches

- Each time step asks a question (query) and compares it with every past step (key):
$$\text{Attention}(Q, K, V) = \text{softmax}\left(QK^{\top} / \sqrt{d_k}\right) V \text{ (Vaswani et al. (2017))}$$
 - ▶ $Q = XW_Q$, $K = XW_K$, $V = XW_V$: linear projections of the inputs X ; softmax turns scores into weights summing to one; $\sqrt{d_k}$ keeps it from saturating
 - ▶ No product of Jacobians over time: any past step is one hop away; cost $O(n^2)$ in the number of inputs n
- **Positional encoding**: attention treats its inputs as a set, so order is added (sinusoids in the original Transformer; rotary embeddings in Chronos-2)
- **Patching** (Nie et al. (2023)): a token is a block of consecutive observations
 - ▶ 512 daily returns in patches of 16 days: 32 tokens, about 256 times fewer attention scores; each token summarises a few weeks of market activity

Patching a 512-Day Context



- The last 512 daily returns of the S&P 500, split as Chronos-2 (16 days) and TimesFM-2.5 (32 days) read them
- Each shaded block becomes one embedding vector for the Transformer
- DeepAR (Salinas et al. (2020)), N-BEATS (Oreshkin et al. (2020)) and PatchTST (Nie et al. (2023)) learn across many series; foundation models push this to its limit

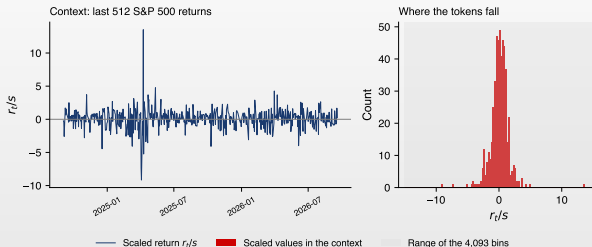
What Is a Foundation Model?

- ▣ A large network pretrained once on a very broad corpus, then used on new tasks
 - ▶ Idea borrowed from large language models (LLM): pretrain on text, then use on any question
- ▣ **Zero-shot** forecasting: no training on the target series; only its recent history (the **context**) is passed at forecast time
 - ▶ Output: a set of quantiles of the future values, not only a point
 - ▶ Fine-tuning (further training on the target series) is possible but not used here
- ▣ Appeal for finance: one model for thousands of series, no estimation per series

Chronos: Time Series as a Language

- ▣ Ansari et al. (2024): turn values into tokens and train a language model (T5) with cross-entropy
 - ▶ Mean scaling: divide the context by $s = \frac{1}{C} \sum_t |x_t|$
 - ▶ Quantisation: 4,093 equally spaced bin centres in $[-15, 15]$ (width 0.0073), plus special tokens
 - ▶ Forecast: sample future tokens, map back to values, read quantiles from the samples
- ▣ Pretraining corpus: public datasets (energy, transport, retail, weather, web, finance and economics) plus synthetic series (TSMixup, KernelSynth)
 - ▶ Model sizes 20 to 710 million parameters; the small version has 46 million

What Chronos Sees: Scaled Returns and Tokens



- Context: the last 512 S&P 500 daily returns; scale $s = 0.672\%$; scaled returns from -9.2 to 13.5, 96% within ± 3
- The 512 returns use 336 distinct tokens out of 4,093: scaling makes the model free of units, so the volatility level must be read from the context
- A crash day M raises s by about $|M|/C$: strongly in a short context, little in a 512-day one (Seminar, A3)

The Three Models We Test

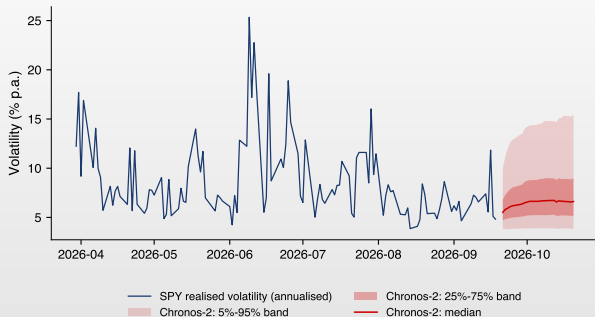
Model	Parameters	Patch	Quantile levels	Weights released
Chronos-Bolt (small)	48 M	16	10%–90%	25.11.2024
TimesFM-2.5	231 M	32	mean, 10%–90%	01.10.2025
Chronos-2	119 M	16	1%, 5%, ..., 99%	30.10.2025

- **Chronos-Bolt:** patches and a direct quantile output instead of token sampling
- **Chronos-2** ([Ansari et al. \(2025\)](#)): robust scaling, attention across time and across groups of related series
- **TimesFM** ([Das et al. \(2024\)](#)): decoder-only Transformer, pretrained on about 10^{11} time points
- All run on a CPU with a 512-day context; the release dates of the weights bound what the models could have seen; only Chronos-2 reaches the 1% level

From a Quantile Grid to VaR and ES

- The model returns quantiles q_u at levels $u_1 < \dots < u_K$
 - ▶ Point forecast: the median $q_{0.5}$, or the mean $\int_0^1 q_u du$; for increasing g , $E[g(Y)] = \int_0^1 g(q_u) du$ (e.g. $g = \exp$ for log variance)
 - ▶ Tail risk (Chapter 7): $\text{VaR}_\alpha = -q_\alpha$, $\text{ES}_\alpha = -\frac{1}{\alpha} \int_0^\alpha q_u du$
- Worked example, S&P 500, day after 18.09.2026: $q_{0.01} = -2.70\%$, $q_{0.05} = -1.40\%$
 - ▶ VaR 1% = 2.70%; linear interpolation $q_{0.025} = q_{0.01} + \frac{0.015}{0.04} (q_{0.05} - q_{0.01}) = -2.21\%$, VaR 2.5% = 2.21%
 - ▶ ES 2.5% = $-\frac{1}{0.025} \left[0.01 q_{0.01} + 0.015 \frac{q_{0.01} + q_{0.025}}{2} \right] = 2.56\%$
- Below the lowest level the grid is silent: holding q_u at $q_{0.01}$ understates ES, interpolation can err either way (Seminar, A5–A6)
 - ▶ Chronos-Bolt and TimesFM stop at 10%: a "VaR 1%" request returns the 10% quantile

A Zero-Shot Fan Chart for Volatility



- Chronos-2 on log realised variance of SPY (512-day context), 22 trading days ahead from 18.09.2026; bands: quantiles 5–95% and 25–75%
- Last volatility 4.8% p.a., 20-day average 6.6%; median forecast at 22 days 6.6%, 90% band 3.8–15.4%
- The band widens and is skewed upwards: mean reversion, as in GARCH and HAR (Chapters 5, 9), with no estimation on SPY
- Whether these forecasts are better is the question of Tests 1–3

 MFM_ch14_fan_chart

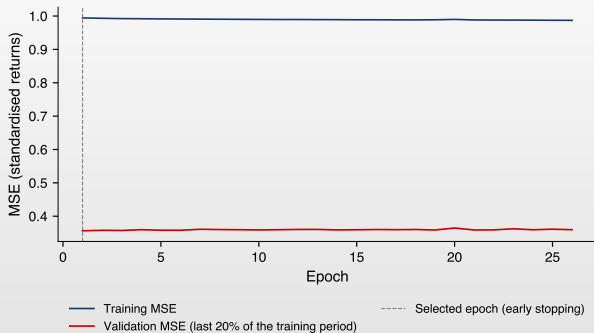
Pretraining Leakage and Contamination

- A test is out of sample only if the test period was never seen during training; four ways it can be “seen”
 - ▶ (i) the same series is in the pretraining corpus; (ii) correlated series carry the same shocks; (iii) developers select model versions on overlapping public benchmarks
 - ▶ (iv) our own look-ahead: scalings, windows, seeds or contexts chosen with test data
- The Chronos corpus includes financial series (e.g. daily exchange rates 1990–2016); we cannot rule out that our series were seen
- Remedy for (i)–(iii): a **post-release window** from 3 November 2025, a common cutoff after all three sets of weights were published
 - ▶ Clean but short (about 220 trading days; 320 calendar days for Bitcoin): low power, especially for VaR 1%
- (iv) is under our control: every scaling, window and estimate uses only past data; a formal protocol is in the [Appendix](#)

Design of the Return Test

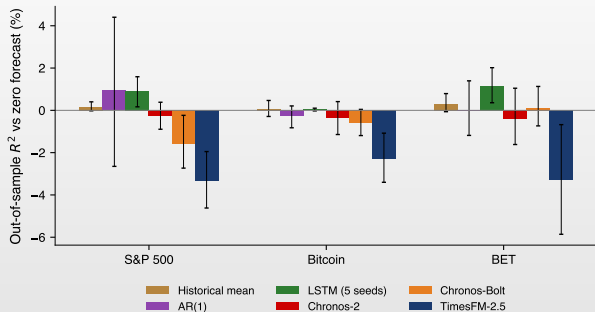
- Daily log returns (%): S&P 500 and BET from 04.01.2016 (2 693 and 2 675 days), Bitcoin from 01.01.2019 (2 818 days)
 - ▶ Every forecast uses only data up to the previous close
- Models: historical mean, AR(1) on a 1,000-day window, LSTM, Chronos-2, Chronos-Bolt and TimesFM-2.5 (median)
 - ▶ Squared error targets the mean; grid means ($\int_0^1 q_u du$) do worse still, e.g. S&P 500 R^2 -0.77% (Chronos-2), -3.42% (TimesFM-2.5)
- Benchmark: the zero forecast; out-of-sample $R^2 = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t r_t^2$
 - ▶ Zero rather than the historical mean, as in [Gu, Kelly & Xiu \(2020\)](#); compare [Campbell & Thompson \(2008\)](#)
 - ▶ Inference: block-bootstrap CI for R^2 (blocks of 20 days), Diebold–Mariano (DM) test on squared errors with Newey–West variance ([Diebold & Mariano \(1995\)](#))

The LSTM for Returns and Early Stopping



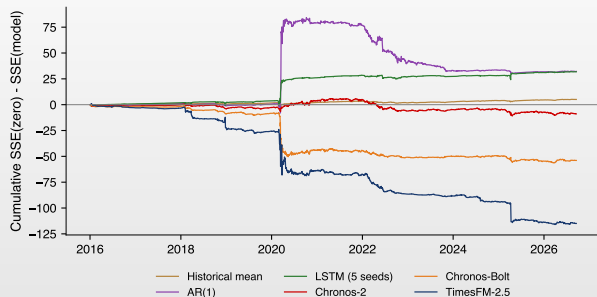
- Input: the last 20 returns, standardised on the training period; 16 hidden units, linear output: $4(hd + h^2 + 2h) + h + 1 = 1233$ parameters
- MSE, Adam, weight decay 10^{-4} ; trained once on up to 4,000 days, last 20% for early stopping; forecast = average of 5 seeds
- Chart (S&P 500, seed 0): training stops 26 epochs in, the weights of epoch 1 are kept
- Little to learn: forecast s.d. 0.047% against 1.13% for returns; validation MSE is lower only because those years were calmer

Out-of-Sample R^2 for Returns



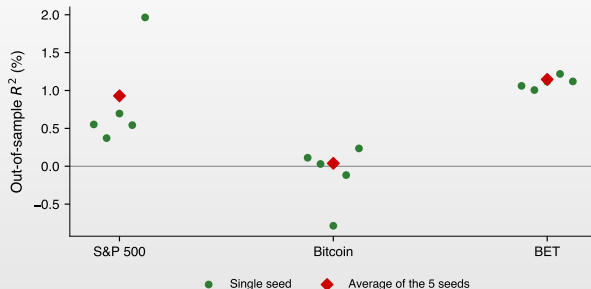
- R^2 against the zero forecast (%), 95% block-bootstrap CIs; full table with DM p -values in the [Appendix](#)
- S&P 500 and Bitcoin: no model beats zero significantly
- TimesFM-2.5 is significantly **worse** than zero (S&P 500 R^2 -3.35%, DM $p = 0.001$): forecast s.d. 0.141%, three times the LSTM's
- BET: LSTM R^2 +1.15% (DM $p = 0.014$): a small signal, tested for nesting, multiplicity and stale prices below

Cumulative Squared-Error Gains on the S&P 500



- $\sum_{s \leq t} [r_s^2 - (r_s - \hat{r}_s)^2]$: rises when the model beats the zero forecast, falls when it loses
- All curves stay near zero in calm years and jump in March 2020: one month decides the ranking
- AR(1) gains in 2020 from the negative autocorrelation of crash days, then gives part back; TimesFM-2.5 loses steadily
- Report the path, not only the final R^2

The Seed Lottery



- Out-of-sample R^2 of five LSTMs that differ only in the random seed, and of their average
- Bitcoin: single seeds from -0.79% to +0.24%; S&P 500 from +0.37% to +1.96%: variation as large as the effect
- BET: all 5 seeds between +1.01% and +1.22%: robust across seeds, not yet robust to multiplicity or stale prices
- Choosing the best seed after seeing the test is backtest overfitting (Chapter 13)

Nested Models and Many Tests

- Zero is nested in the historical mean and AR(1): under H_0 the larger model only adds estimation noise, so DM rejects too rarely
 - ▶ Clark & West (2007): $\tilde{d}_t = e_{0,t}^2 - [e_{1,t}^2 - (\hat{r}_{0,t} - \hat{r}_{1,t})^2]$, one-sided HAC t -test
 - ▶ One-sided p , DM vs Clark–West: historical mean 0.048 vs 0.007 (S&P 500), 0.041 vs 0.003 (BET); AR(1) 0.320 vs 0.055
- 6 models $\times$ 3 markets = 18 tests; the smallest one-sided DM p is the BET LSTM, 0.007
 - ▶ Bonferroni threshold 0.0028; Holm-adjusted p over the 18 tests 0.130: not significant
 - ▶ Within the six BET models: Romano–Wolf (Romano & Wolf (2005)) 0.016; SPA (Hansen (2005)), H_0 : no model beats zero: BET 0.018, S&P 500 0.130, Bitcoin 0.354
- Reading: “some model beats zero on the BET” survives; “one of 18 comparisons is real” does not. Fix the family before looking

Is BET Predictability Real? Non-Synchronous Trading

- Stale prices of thinly traded constituents make index returns positively autocorrelated, even if every price is efficient (Lo & MacKinlay (1990))
 - ▶ First-order autocorrelation since 2016: BET 0.06, S&P 500 -0.14
 - ▶ The LSTM forecast has correlation 0.88 with the previous day's BET return: it mostly learned this autocorrelation
- Tradable check: the BET-tracking ETF Patria-TVBETETF, 2629 common days, same LSTM signal
 - ▶ Same day: R^2 2.88% (BET 1.26%), but the ETF is thinly traded too (autocorrelation 0.07, 4.2% zero-return days)
 - ▶ Executed one day later: R^2 0.57%, Clark–West $p = 0.003$
- Reading: the evidence is consistent with stale-price effects; a small signal survives the execution lag, before transaction costs (Chapter 2)

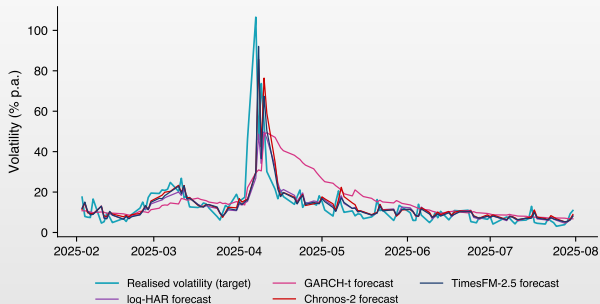
Target, Benchmark and Loss

- ▣ **Realised variance** (RV, [Andersen et al. \(2003\)](#)): sum of squared 5-minute log returns within the trading day
 - ▶ $RV_t = \sum_{j=1}^{78} r_{t,j}^2$ for SPY (regular session, 6.5 hours); average volatility 11.9% p.a.
- ▣ **HAR** ([Corsi \(2009\)](#)): $RV_t = \beta_0 + \beta_d RV_{t-1} + \beta_w \overline{RV}_{t-5:t-1} + \beta_m \overline{RV}_{t-22:t-1} + \varepsilon_t$
 - ▶ log-HAR: the same on log RV, forecast $\exp(\hat{y} + \hat{s}^2/2)$: the mean if the log error is conditionally Normal
- ▣ **QLIKE** loss ([Patton \(2011\)](#)): $L = RV_t/\hat{h}_t - \log(RV_t/\hat{h}_t) - 1$
 - ▶ Minimised by the conditional mean; ranks forecasts as the true variance would when RV is a conditionally unbiased proxy, $E[RV_t | \mathcal{F}_{t-1}] = \sigma_t^2$ (so does MSE)

Design of the Volatility Test

- ▣ SPY realised variance, one-day-ahead forecasts from 11.10.2022 to 18.09.2026 (979 days)
 - ▶ HAR and log-HAR on a rolling 500-day window; random walk: $\hat{h}_t = \text{RV}_{t-1}$
 - ▶ GARCH-t on daily SPY returns, rescaled to RV units by the ratio of average RV to average squared return over 500 days
 - ▶ LSTM on the last 22 values of log RV, retrained every 125 days on 500 days (hidden size 8, three seeds)
- ▣ Foundation models on log RV (context 512): forecast $\hat{h}_t = \int_0^1 \exp(q_u) du$ on the quantile grid
 - ▶ Post-release window: 218 days from 3 November 2025

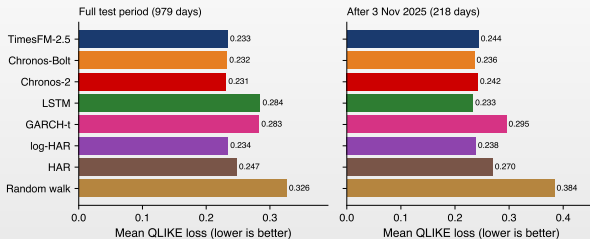
Forecasts Through the April 2025 Shock



- SPY realised volatility (annualised) and one-day-ahead forecasts, February–July 2025; peak 107% p.a. on 07.04.2025, after the US tariff announcements
- No model anticipates the jump; all react the next day: one-day forecasts extrapolate the past, shocks come from news
- Foundation models and log-HAR decay back at similar speeds; GARCH-t reacts to daily returns and decays more slowly

 MFM_ch14_realised_vol

QLIKE Losses of Eight Volatility Models



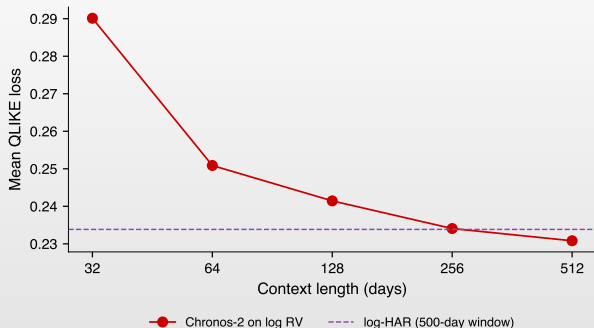
- Left: full test period; right: after 3 November 2025 (218 days); full table with DM and MCS p -values in the [Appendix](#)
- Chronos-2 (0.231), Chronos-Bolt, TimesFM-2.5 and log-HAR (0.234): DM tests not significant (Chronos-2 $p = 0.537$), all in the 90% MCS
- Random walk and GARCH-t are eliminated; HAR in levels over-forecasts (mean forecast / mean RV = 1.36); after release the small LSTM is best (0.233): rankings are unstable
- No conditional difference: Giacomini–White test ($1, d_{t-1}$) against log-HAR, $p = 0.760$ (Chronos-2), 0.935 (TimesFM-2.5); no difference detected is not proof of equality

Median or Mean? A Costly Detail

Chronos-2 and TimesFM-2.5 on log RV	QLIKE	Mean forecast / mean RV
Chronos-2, $\int_0^1 \exp(q_u) du$	0.231	1.02
Chronos-2, $\exp(q_{0.5})$	0.259	0.79
TimesFM-2.5, $\int_0^1 \exp(q_u) du$	0.233	0.94
TimesFM-2.5, $\exp(q_{0.5})$	0.259	0.79

- A theorem, not a curiosity ([Gneiting \(2011\)](#)): QLIKE is consistent for the mean, $\arg \min_h E[L(RV, h) | \mathcal{F}] = E[RV | \mathcal{F}]$
 - ▶ Monotone maps preserve quantiles: $\exp(\text{median of log RV})$ is the median of RV, elicited by absolute error, not by QLIKE
 - ▶ Lognormal RV: median/mean = $e^{-s^2/2}$; the median version is significantly worse (DM +3.60 for Chronos-2) and worse than log-HAR (0.234)

How Much Context Does Chronos-2 Need?

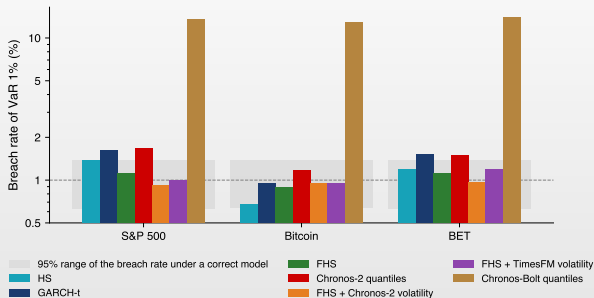


- QLIKE of Chronos-2 on log RV with contexts of 32 to 512 days, on the same forecast days
- QLIKE falls from 0.290 (32 days) to 0.231 (512 days); below about 256 days Chronos-2 is worse than log-HAR (0.234)
- Long memory: months of history reveal the slow-moving level, which HAR encodes explicitly through its 22-day average
- Context length is a modelling choice to report and test, like a window length

Design of the Tail-Risk Test and the FZ0 Loss

- $\text{VaR}_\alpha = -q_\alpha(r_{t+1} \mid \mathcal{F}_t)$, the loss exceeded with probability α ; $\text{ES}_\alpha = -\frac{1}{\alpha} \int_0^\alpha q_u du$; levels VaR 1% (coverage tests) and (VaR 2.5%, ES 2.5%)
- Classical models, 1,000-day window, re-estimated every 20 days: HS, GARCH(1,1)-t, FHS (McNeil & Frey (2000))
 - ▶ **Raw**: Chronos-2 quantiles of tomorrow's return (context: 512 returns)
 - ▶ **Hybrid**: scale $\hat{\sigma}_t$ = forecast of $|r_t|$ (Chronos-2: grid mean; TimesFM: median); FHS on $r_t/\hat{\sigma}_t$
- ES alone is not elicitable, the pair (VaR, ES) is (Fissler & Ziegel (2016)); FZ0 (Patton, Ziegel & Chen (2019)), loss $L_t = -r_t$, $v = \text{VaR}$, $e = \text{ES} > 0$:
 - ▶ $\text{FZ0} = \frac{1\{L_t > v\}(L_t - v)}{\alpha e} + \frac{v}{e} + \log e - 1$
 - ▶ Lower is better: penalises breaches (first term) and wide forecasts ($\log e$); models compared by DM tests on FZ0 differences and by the MCS (Hansen, Lunde & Nason (2011))

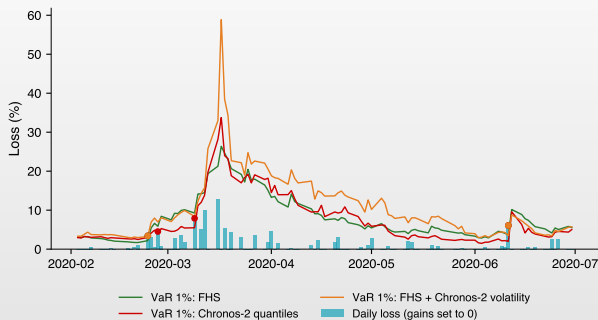
VaR 1% Breach Rates



- Share of days with a loss above VaR 1% (log scale); grey band: 95% range if the model were correct
- Raw Chronos-2: too many breaches on the S&P 500 (1.67%, Kupiec $p = 0.001$) and the BET (1.50%)
- Chronos-Bolt, lowest quantile 10% used as "VaR 1%": 13.5%
- Hybrids: breach rates close to 1% in all three markets (average coverage only; full table in the

► [Appendix](#)

The COVID-19 Crash: VaR 1% of Three Models



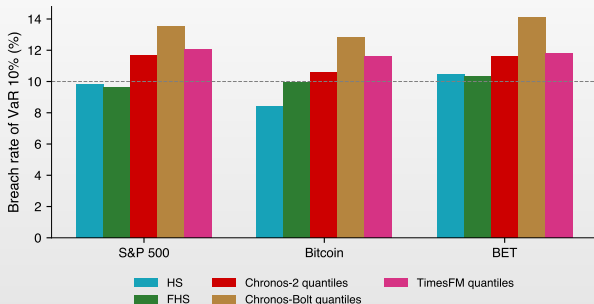
- S&P 500, February–June 2020 (104 days): daily losses and VaR 1% forecasts; dots: breaches
- Breaches: FHS 2, raw Chronos-2 4, FHS with Chronos-2 volatility 2, HS 12; all models are hit in the first days
- Chronos-2 reads the context fast, so its quantiles widen quickly
- The hybrid overshoots: VaR 1% of 58.9% on 17.03.2020, against 26.4% for FHS and a largest loss of 12.8% ([► Appendix](#))

Backtesting Beyond Kupiec: Clustering and the DQ Test

p : indep. / DQ	HS	GARCH-t	FHS	Chronos-2 raw	FHS + Chronos-2	FHS + TimesFM
S&P 500	0.001 / <0.001	0.205 / 0.001	0.045 / 0.021	0.223 / <0.001	0.021 / 0.001	0.029 / <0.001
Bitcoin	0.611 / 0.760	0.261 / 0.463	0.220 / 0.092	0.404 / 0.034	0.026 / 0.004	0.261 / 0.182
BET	0.399 / <0.001	0.258 / 0.082	0.409 / 0.650	0.270 / 0.026	0.475 / <0.001	0.379 / <0.001

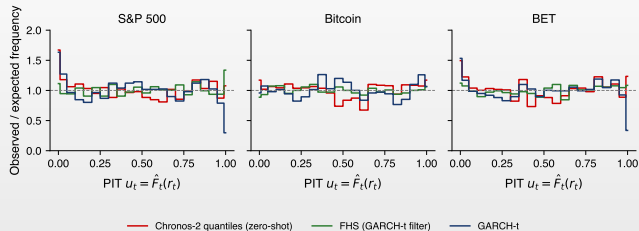
- Independence ([Christoffersen \(1998\)](#)): first-order Markov chain of breaches, $LR \sim \chi^2_1$; conditional coverage = Kupiec + independence, χ^2_2
- Dynamic quantile (DQ, [Engle & Manganelli \(2004\)](#)): regress $I_t - \alpha$ on a constant, four lagged hits and VaR_t ; Wald statistic $\sim \chi^2_6$
- S&P 500: the hybrids have the right breach rate but fail independence and DQ; every model fails DQ at 5%
 - BET: FHS passes both tests, the hybrids fail DQ; right on average is not right conditionally

Where Deciles Suffice: VaR 10%



- Breach rate of the 10% quantile (target 10%) for HS, FHS and the three foundation models
- S&P 500: HS 9.8%, FHS 9.6%; Chronos-2 11.7%, Chronos-Bolt 13.5%, TimesFM-2.5 12.0%
- Kupiec rejects the foundation models in eight of nine cases: the lower tail sits too high, through location, scale or tail shape
- Plausible reason: pretraining corpora are dominated by smoother series; as volatility filters in FHS the models inherit the tails from the data

Calibration of the Whole Distribution: the PIT



- PIT (probability integral transform) $u_t = \hat{F}_t(r_t)$: i.i.d. uniform if the predictive distribution is correct (Diebold, Gunther & Tay (1998)); bars: observed over expected frequency in the 22 intervals of the Chronos-2 grid
- Berkowitz test (Berkowitz (2001)) of $\mu = 0, \sigma = 1$ for $\Phi^{-1}(u_t)$, censored outside [1%, 99%]: Chronos-2 rejected on the S&P 500 ($p = 0.019, \hat{\sigma} = 1.03$) and the BET; FHS never
- Quantile score $QS = \frac{2}{K} \sum_k (1\{r_t < q_{u_k}\} - u_k)(q_{u_k} - r_t)$, a grid analogue of the CRPS (Gneiting & Raftery (2007)):
Chronos-2 worse than FHS in the centre and the tails (S&P 500 DM +2.51, +3.65); table in the [Appendix](#)

Comparing (VaR 2.5%, ES 2.5%) Forecasts: the MCS

HS	0.140	0.343	0.015
GARCH-t	0.187	0.597	0.341
FHS	1.000	0.597	1.000
Chronos-2 quantiles	0.140	0.641	0.239
FHS + Chronos-2 volatility	0.140	1.000	0.341
FHS + TimesFM volatility	0.183	0.704	0.202
	S&P 500	Bitcoin	BET

- p -values of the 90% model confidence set (Hansen, Lunde & Nason (2011)) for the FZ0 loss; green: kept, red: eliminated; FZ0 table in the [Appendix](#)
- S&P 500: FHS has the lowest FZ0; the hybrids have the right breach rate but clustered breaches and a larger VaR (3.20% vs 2.78%)
- The MCS eliminates only HS on the BET; pairwise DM is sharper: FHS beats the Chronos-2 hybrid on the S&P 500 (DM +2.65, $p = 0.008$) and the BET (DM +2.36)

After the Release: Few Breaches, Little Power

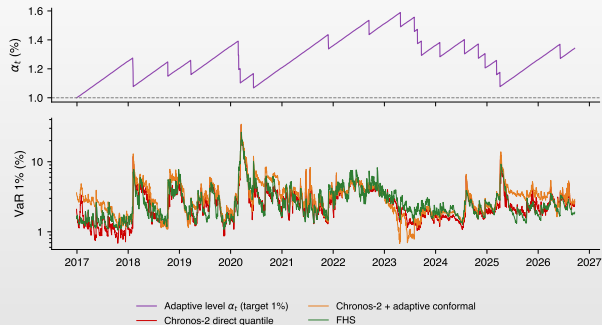
Breaches after 3.11.2025	HS	GARCH-t	FHS	Chronos-2 quantiles	FHS + Chronos-2 vol.	FHS + TimesFM vol.
S&P 500 (220 days)	0	2	2	2	2	2
Bitcoin (320 days)	3	3	4	4	3	4
BET (217 days)	4	4	4	5	1	2

- Expected breaches: 2.2 (S&P 500), 3.2 (Bitcoin), 2.2 (BET): no contamination is possible, but the tests have little power
 - ▶ Exact two-sided Kupiec test at 5% against a true rate of 1.67%: power 0.19 at $T = 220$, actual size 0.13; 0.50 at $T = 1,000$; 0.87 at $T = 2,693$
 - ▶ 80% power first at about 2 130 days (8.4 years), for good from 2 320; against 2%: power 0.29 at $T = 220$ (Seminar A8–A9)
- Low power is not zero power: HS has 0 breaches on the S&P 500, and $x = 0$ is in the Kupiec rejection region at $T = 220$; a non-rejection does not establish calibration

Recalibrating Quantiles: Adaptive Conformal Inference

- Score $E_t = \hat{q}_{\alpha,t} - r_t$; corrected quantile $\hat{q}_{\alpha,t} - \hat{Q}_{1-\alpha_t}$, with $\hat{Q}$ the empirical $(1 - \alpha_t)$ quantile of the last 250 scores
- Adaptive level (Gibbs & Candès (2021)): $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$, $\text{err}_t = 1\{r_t < \hat{q}_{\alpha,t} - \hat{Q}_{1-\alpha_t}\}$, $\alpha_1 = \alpha$, $\gamma = 0.001$;
 $\gamma = 0$: plain split conformal
 - After a breach α_t falls (the next VaR is wider); after a calm day it rises slowly
- Guarantee for any sequence, no model needed: $|\frac{1}{T} \sum_t \text{err}_t - \alpha| \leq \frac{\max(\alpha_1, 1 - \alpha_1) + \gamma}{\gamma T}$, a long-run statement only
- ES 2.5%: Chronos-2 ES shifted by the VaR 2.5% correction, a heuristic without a conformal guarantee; rules and the full table in the [Appendix](#)

Adaptive Conformal VaR 1% on the S&P 500



- Top: adaptive level α_t (target 1%); bottom: VaR 1% of raw Chronos-2, Chronos-2 with ACI (adaptive conformal inference), FHS (log scale)
- ACI brings the breaches near 1% (S&P 500: 21 vs 24.4 expected); conditional coverage is not rejected in any market
- DQ still rejects on the S&P 500 and the BET; the VaR is wider (3.36% vs 2.85% for FHS) and FZ0 significantly higher
- Proper scoring rules reward sharpness subject to calibration ([Gneiting & Raftery \(2007\)](#)): calibration is necessary, not sufficient

Case Study: Is Complexity a Virtue in Return Prediction?

- **Question** (Kelly, Malamud & Zhou (2024)): must a return-prediction model have far fewer parameters P than observations T , or can models with complexity $c = P/T > 1$ do better out of sample?
 - ▶ Setting (eqs. 1–2): $R_{t+1} = f(G_t) + \varepsilon_{t+1}$, f unknown, approximated by a wide one-hidden-layer network $\sum_{i=1}^P S_{i,t} \beta_i$ (universal approximation, Hornik, Stinchcombe & White (1989))
- **Link to Test 1**: on the S&P 500 and Bitcoin no model beats the zero forecast significantly; the paper asks whether a negative out-of-sample R^2 means no economic value
 - ▶ Economic value: Sharpe ratio of the market-timing return $\pi_t R_{t+1}$, with position $\pi_t = \hat{\beta}' S_t$ (eq. 6)
- **Theory** (Sections III–IV): under the assumptions of the paper, a misspecified model with optimally chosen ridge shrinkage has expected out-of-sample R^2 and timing Sharpe ratio rising with c
 - ▶ Near $c = 1$ (the interpolation boundary) least squares fits the training data exactly and its variance explodes; beyond it, performance recovers
- **Why it matters**: a theoretical case for the large models of this chapter, against the parsimony rule of classical econometrics

Case Study: Data and Method

- **Data** (Section V.A): monthly excess return of the CRSP value-weighted index; the 15 predictors of [Welch & Goyal \(2008\)](#) (14 variables and the lagged market return), 1926–2020
 - ▶ Returns scaled by their trailing 12-month volatility, predictors by an expanding-window standard deviation; 36 months of warm-up, so the sample starts in 1930
- **Random Fourier features** (eq. 20, [Rahimi & Recht \(2007\)](#)): $S_{i,t} = (\sin(\gamma \omega_i' G_t), \cos(\gamma \omega_i' G_t))'$, $\omega_i \sim N(0, I)$, $\gamma = 2$
 - ▶ A wide two-layer network whose first-layer weights ω_i are drawn at random and fixed; only the output weights β are estimated
- **Ridge** (Section II.A): $\hat{\beta}(z) = (zI + T^{-1} \sum_t S_t S_t')^{-1} T^{-1} \sum_t S_t R_{t+1}$, no intercept; $z \rightarrow 0^+$: the ridgeless (pseudo-inverse) solution
- **Design** (Section V.C): rolling windows of $T = 12, 60, 120$ months; P from 2 to 12 000; $\log_{10} z = -3, \dots, 3$; results averaged over 1 000 independent draws of the features
- **Evaluation**: out-of-sample $R^2 = 1 - \text{Var}(\text{error})/\text{Var}(R)$ (footnote 40; the theory divides the MSE by $E[R^2]$, the zero forecast of Test 1); annualised Sharpe ratio; alpha and information ratio (alpha over residual volatility) from a regression of the timing return on the market
 - ▶ Benchmark: the linear "kitchen sink" regression on all 15 predictors (Section V.D)

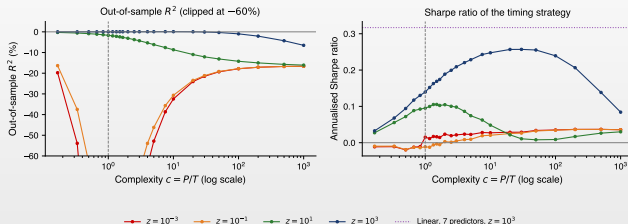
Case Study: Key Results

Model ($T = 12$)	R^2 (%)	Sharpe [t]	Inf. ratio [t]	Max loss	Skewness
Linear, $z = 0^+$	< -100	$-0.11 [-1.0]$	$-0.16 [-1.6]$	98.5	-0.9
Linear, $z = 10^3$	-3.8	$0.46 [4.4]$	$0.33 [3.1]$	2.4	-0.1
RFF, $c = 1000$, $z = 10^3$	0.6	$0.47 [4.5]$	$0.31 [2.9]$	1.2	2.5

Source: Table I, Panel A, of the paper; information ratio against the market; maximum monthly loss in standard deviations

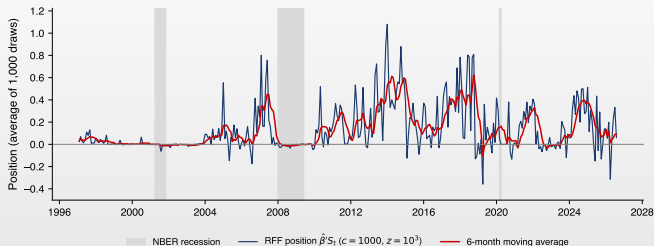
- ▣ Ridgeless kitchen sink ($P = 15 > T = 12$): R^2 below -100% , Sharpe ratio -0.11
 - ▶ Reading: at the interpolation boundary forecasts explode, reproducing the negative verdict of [Welch & Goyal \(2008\)](#)
- ▣ Same predictors with $z = 10^3$: $R^2 = -3.8\%$, yet Sharpe ratio 0.46 ($t = 4.4$)
 - ▶ Reading: R^2 penalises the scale of forecasts; timing profits depend on their correlation with returns (footnote 5)
- ▣ RFF, $c = 1000$: information ratio 0.26 ($t = 2.5$) against the best linear model; with $T = 60$ and 120 , 0.25 and 0.24 against the market ($t = 2.3$ and 2.2 ; Panels B–C)
 - ▶ Reading: nonlinearity extracts more from the same predictors, with a smaller maximum loss and positive skewness
- ▣ Positions (Figure 10): almost always long; cut before 14 of 15 NBER recessions, out of sample
 - ▶ Reading: the model learns the sign restriction of [Campbell & Thompson \(2008\)](#) without being given it

Case Study: Complexity on Course Data



- Figures 7–8 of Kelly, Malamud & Zhou (2024) on course data: SPY excess returns, 7 predictors, 354 test months (February 1997–August 2026); $T = 12$, $\gamma = 2$, 1 000 draws; dashed: $c = 1$; dotted: linear, $z = 10^3$
- Nearly ridgeless ($z = 10^{-3}$): $R^2 = -1842\%$ at $c = 1$, recovering to -16.6% at $c = 1\,000$: the double descent of Figure 7 (left panel cut at -60%)
- $z = 10^3$: Sharpe ratio 0.14 at $c = 1$, 0.26 at $c = 30$, 0.08 at $c = 1\,000$ ($t = 0.5$); linear with $z = 10^3$: Sharpe 0.32 ($t = 1.7$), not beaten by RFF (information ratio -0.17)
- Reading: complexity helps up to $c \approx 30$, not up to $c = 1\,000$, with 7 of the 15 predictors and 29.5 years

Case Study: Timing Positions and Recessions



- Figure 10 of [Kelly, Malamud & Zhou \(2024\)](#) on course data: position $\hat{\beta}' S_t$ of RFF with $c = 1000$, $z = 10^3$; shaded: NBER recessions (FRED USREC); red: 6-month moving average
- Long in 72% of months, average position 0.13: a positive equity premium learned without a sign restriction
- Average position 0.07 in the 6 months before the 3 recessions and -0.01 during them, against 0.15 otherwise; the 2020 pandemic shock is not anticipated
- Three recessions are a description, not a test: the paper has 15 recessions and 90 years

Case Study: Limits and an Open Question

- **Limits** stated by the authors
 - ▶ Theory for linear ridge models on fixed random features and a single asset; a panel with cross-asset covariances is left open (Introduction, footnote 2)
 - ▶ Predictability spread evenly across features (Assumption 4); sparse LASSO-type models left to future research (footnote 22)
 - ▶ No optimal degree of nonlinearity γ ; average timing returns about half as large in 1975–2020 as in 1930–1974 (Section V.F)
- **Course data** (Seminar C3): 7 of the 15 predictors and about 30 years; the power of a two-sided 5% test for an information ratio of 0.31 is only about 39% ($t \approx 0.31\sqrt{29.5} \approx 1.7$)
- **Open question**: learned instead of random features, the embeddings of a pretrained foundation model such as Chronos-2 as S_t ; is complexity still a virtue, on a panel of course markets?

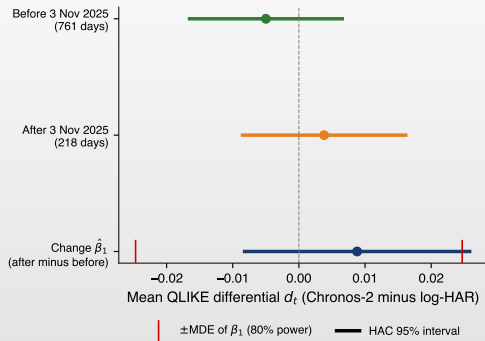
An Open Question: Skill or Memory?

- Test 2: zero-shot Chronos-2 is not distinguishable from log-HAR on SPY realised variance (QLIKE 0.231 vs 0.234, DM $p = 0.537$)
- Rahimikia, Ni & Wang (2025): on daily excess returns across global markets, off-the-shelf foundation models do poorly zero-shot and after fine-tuning; models pretrained from scratch on financial data gain
- Lopez-Lira, Tang & Zhu (2025): once a model has seen the realised values, its counterfactual forecasting skill is not identified; detected memorisation is only a lower bound (shown for LLMs; the logic applies to any pretrained forecaster)
- Open: is the parity learned structure (long memory, mean reversion) or recall of the test period, and can a short post-release window tell?
 - $d_t = L(\hat{h}_t^{FM}, RV_t) - L(\hat{h}_t^{\log HAR}, RV_t)$; memorisation implies $E[d_t | \text{post}] > E[d_t | \text{pre}]$; H_0 : equality; a rejection may also reflect a regime change

The Discovery Loop in Five Steps

- **Literature:** ask an assistant for evaluations of time-series foundation models on financial data since 2024; check every arXiv identifier and read the corpus documentation yourself
- **Hypotheses:** predictions that separate memorisation from skill: pre vs post release, never-public series, simulated series with the same dynamics
- **Data and code:** from the chapter's volatility forecasts, the QLIKE differential Chronos-2 minus log-HAR, a pre/post test with HAC errors and the MDE; first reproduce the log-HAR QLIKE, 0.234
- **Robustness,** fixed in advance: a placebo release date one year earlier must show no break; simulated series cannot be memorised
- **Critique:** ask the assistant to attack "no memorisation, because the post-release test is insignificant"; expect low power and different regimes (reusable prompts in the [▶ Appendix](#))

Mini-Case: a Test Without Power



- Typical assistant proposal: “the parity of Chronos-2 comes from memorised data, so its edge must vanish after release”
- Mean d_t : -0.0050 before 3 Nov 2025 (761 days), +0.0038 after (218 days); change +0.0088, HAC $t = 1.00$, $p = 0.32$
- $MDE = (z_{0.975} + z_{0.8}) se_{HAC}(\hat{\beta}_1) = 0.0247$: 4.9 times the pre-release gap; with an infinite post window still 0.0169
- Reading: “no memorisation” is not a finding; power must come from the cross-section, not from waiting

Pitfalls the Researcher Must Check

- **Leakage:** pretraining overlap, full-sample scaling, HAR averages that include day t (Seminar B8 cuts QLIKE by a quarter)
 - ▶ Test: post-release window, simulated or private series; recompute the forecast for t from data truncated at $t - 1$
- **Weak benchmarks:** compare with zero (returns), log-HAR (volatility), FHS (tails), not with a random walk
- **Wrong functional:** medians used as means (mean forecast / mean RV should be near 1; Chronos-2 median: 0.79); deciles used as 1% quantiles
- **Selection:** seeds, contexts and models chosen after seeing the test; log every trial, compare the whole set with the MCS
- **Wrong inference:** DM on nested models, no multiplicity control, coverage without independence, “not significant” reported without the MDE

Your Turn: a Project Seed

- Question: does zero-shot skill for daily variance survive where memorisation is impossible: after release, on simulated series, on less-covered markets?
- Replicate first: the chapter's QLIKE of log-HAR (0.234) and Chronos-2 (0.231), DM $p = 0.537$, and the 218-day post-release window
- Deliverables
 - ▶ A pre-registered power calculation; a panel of series from the course data with a range-based variance proxy and a proxy-robust loss ([Patton \(2011\)](#))
 - ▶ Pre/post tests with placebo release dates; the MCS across models; a referee-style critique and your reply
- Log each prompt that shaped a decision in AI_USE.md; each invented reference, leak or wrong number caught in AI_ERRORS.md, with the test that caught it ([templates](#))

When to Use What

Target	Reference model	Where deep learning can help
Daily index returns	Zero forecast	Less efficient markets (BET); many assets, many features
Realised volatility	log-HAR	Zero-shot forecasts for thousands of series without estimation
VaR 1%, ES 2.5%	FHS (GARCH filter)	Volatility filter in FHS or conformal recalibration

- Foundation models are cheap to try: no estimation, one line of code, a CPU is enough
- Their value is breadth (many series, no tuning), not accuracy on a single well-studied series

Key Takeaways

- ▣ LSTM gates can mitigate vanishing gradients (with a forget-gate bias of 0 they still vanish); attention connects distant time steps directly
- ▣ Foundation models forecast zero-shot from a context window and return quantiles
- ▣ Returns: nothing beats zero on large markets; volatility: foundation models not distinguishable from log-HAR; tails: use them only as volatility filters
- ▣ Evaluate every new model against a strong benchmark and after the release of its weights, with inference that matches the design: Clark–West for nested models, SPA or Romano–Wolf for many models, PIT, quantile scores and DQ for distributions

Check Yourself, and Next: Chapter 15

Retrieval questions

- Why does a stable RNN forget at the rate at which a VAR(1) impulse response decays?
- Why is $\exp(q_{0.5})$ of log RV a biased forecast under QLIKE?
- Why can a hybrid VaR pass Kupiec and still lose the FZ0 comparison?
- Why is “not significant after release” not evidence against memorisation?

Chapter 15: LLMs and sentiment analysis

- Can text read by a large language model forecast returns or volatility?
- The same threats return: look-ahead, memorisation of the test period, weak benchmarks
- The same tools: post-release windows, Clark–West, multiplicity control

Appendix: Norms, Spectral Radius and Explosion

- $\gamma_\varphi \|W\|_2 < 1$ is sufficient for geometric vanishing, so $\gamma_\varphi \|W\|_2 \geq 1$ is necessary for explosion (Pascanu, Mikolov & Bengio (2013))
- Spectral radius ρ : in a linear RNN, $\rho(W) > 1$ is necessary for exponential explosion ($\rho(W) = 1$ with a Jordan block still gives polynomial growth)
 - ▶ With φ the local Jacobians $D_t W$ matter: $\rho(W) = 0$ yet $\rho(DW) > 1$ is possible
- Stable VAR: impulse responses and gradients decay like $\rho(A)^k$ up to a factor k^{m-1} if A has a Jordan block of size m
- DUx_t is an innovation only if the inputs x_t are centred white noise; long-memory inputs would pass through the network

Appendix: Geometric versus Hyperbolic Decay, Fitted

- Constant gates, i.i.d. candidate input g_t , stationary start: $\text{Corr}(c_t, c_{t-k}) = f^k$; ARFIMA with $0 < d < 1/2$: $\rho_k \sim Ck^{2d-1}$
- S&P 500 absolute returns since 2000: ACF 0.289 at lag 1, 0.132 at lag 50
 - ▶ Geometric decay with its own amplitude, $\rho_k = \rho_1 f^{k-1}$ (an AR(1) signal observed with noise), matches both lags with $f = 0.984$: $b_f = 4.1$, half-life 43 days
 - ▶ That f implies 0.027 at lag 150; the data show 0.059, more than twice as much (95% band ± 0.024)
- A constant gate $f < 1$ is a short-memory filter; a long context or explicit long averages, as in HAR (Test 2), only approximate long memory over a finite range of lags

Appendix: Other Foundation Models and Their Corpora

- Chronos-Bolt: quantile levels 10%, 20%, . . . , 90% only; Chronos-2: training corpus of selected Chronos and [GIFT-Eval](#) datasets plus synthetic multivariate series; benchmarks GIFT-Eval ([Aksu et al. \(2024\)](#)) and fev-bench
- TimesFM corpus: Google Trends, Wikipedia page views, synthetic series, M4, electricity, traffic, weather; TimesFM-2.5 returns the mean plus deciles
 - ▶ Release dates of the weights, from the Hugging Face repositories: TimesFM-2.5, updated weight version announced on 15.09.2025, released on 01.10.2025; Chronos-2 announced on 20.10.2025, released on 30.10.2025
- **Moirai** ([Woo et al. \(2024\)](#)): any number of variables, mixture-distribution output; Moirai 2.0 ([Liu et al. \(2025\)](#)); non-commercial licence, not used in our tests
- **Lag-Llama** ([Rasul et al. \(2023\)](#)): decoder-only model with lagged values as features and a Student- t output

Appendix: A Formal Contamination Protocol

- Pre/post regression on the loss differential $d_t = L_t^{FM} - L_t^{bench}$: $d_t = \beta_0 + \beta_1 \mathbf{1}\{t \geq \tau\} + u_t$, HAC s.e.
 - ▶ τ : release date of the weights (a common cutoff when several models are compared); memorisation predicts $\beta_1 > 0$; H_0 : $\beta_1 = 0$
 - ▶ Report the minimum detectable effect (MDE) of β_1 : with 218 post-release days it is 4.9 times the pre-release gap (AI section)
- Pre-registration: forecasts submitted before the outcome exists make contamination impossible by design
 - ▶ Live benchmarks: TS-Arena ([Meyer et al. \(2025\)](#)), Impermanent ([Garza et al. \(2026\)](#)); our post-release window is a manual version
- Ablations: removing the language-model part of LLM-based forecasters does not degrade accuracy, often improves it ([Tan et al. \(2024\)](#))
- Electricity prices: [Pan & Ezzat \(2026\)](#) benchmark on two datasets to limit contamination; foundation models are competitive but do not consistently beat domain-specific methods

Appendix: Return Forecasts, Full Table

R^2 (%) [DM p]	Historical mean	AR(1)	LSTM (5 seeds)	Chronos-2	Chronos-Bolt	TimesFM-2.5
S&P 500	+0.15 [0.095]	+0.94 [0.639]	+0.93 [0.090]	-0.26 [0.418]	-1.57 [0.078]	-3.35 [0.001]
Bitcoin	+0.07 [0.691]	-0.28 [0.326]	+0.04 [0.243]	-0.35 [0.413]	-0.61 [0.070]	-2.29 [0.001]
BET	+0.30 [0.081]	-0.00 [1.000]	+1.15 [0.014]	-0.39 [0.562]	+0.12 [0.831]	-3.27 [0.070]

- Squared error targets the mean; grid means ($\int_0^1 q_u du$) do worse still, e.g. S&P 500 R^2 -0.77% (Chronos-2), -3.42% (TimesFM-2.5)
- DM: two-sided test on squared errors against the zero forecast, Newey–West variance

Appendix: Nested Forecasts, the BET AR(1) and Rolling Windows

- BET AR(1), $R^2 = -0.00\%$ and DM $p = 1.000$, is not a degenerate forecast: s.d. 0.074%, average slope 0.046
 - ▶ Squared-error gains of -12.7 in 2020 and +14.7 in 2026 cancel; without these two years $R^2 = -0.11\%$
- Rolling or re-trained models: [Giacomini & White \(2006\)](#) test the forecasting method, estimation window included ([Diebold \(2015\)](#) on the use and abuse of DM)
- SPA refines White's Reality Check ([White \(2000\)](#)): very poor models (TimesFM) no longer dilute the test; both use a stationary bootstrap with mean block 20 days

Appendix: QLIKE, DM and MCS, Full Table

Model	QLIKE	DM vs log-HAR	p	MCS p	QLIKE after 3.11.2025
Random walk	0.326	+4.53	<0.001	0.00	0.384
HAR	0.247	+1.30	0.195	0.29	0.270
log-HAR	0.234	—	—	0.95	0.238
GARCH-t	0.283	+3.43	<0.001	0.03	0.295
LSTM	0.284	+1.68	0.094	0.29	0.233
Chronos-2	0.231	-0.62	0.537	1.00	0.242
Chronos-Bolt	0.232	-0.18	0.857	0.95	0.236
TimesFM-2.5	0.233	-0.08	0.938	0.95	0.244

- Literature: [Brini \(2026\)](#) compares nine zero-shot foundation models with eight HAR-type models on 50 assets; pooled losses favour foundation models because of a few outlier assets, while asset by asset only one small model beats log-HAR at every horizon, narrowly
- [Goel et al. \(2025\)](#): TimesFM zero-shot is a reasonable baseline; incremental fine-tuning is needed to beat econometric models

Appendix: VaR 1% Breaches and Kupiec Tests, Full Table

Breaches [Kupiec p]	HS	GARCH-t	FHS	Chronos-2 quantiles	FHS + Chronos-2 vol.	FHS + TimesFM vol.
S&P 500 (26.9)	37 [0.065]	44 [0.002]	30 [0.559]	45 [0.001]	25 [0.705]	27 [0.989]
Bitcoin (28.2)	19 [0.065]	27 [0.822]	25 [0.539]	33 [0.374]	27 [0.822]	27 [0.822]
BET (26.8)	32 [0.322]	41 [0.010]	30 [0.536]	40 [0.016]	26 [0.884]	32 [0.322]

- In brackets after the market: the expected number of breaches, 1% of the days
- Kupiec: two-sided likelihood-ratio test of a 1% breach rate, χ_1^2

Appendix: COVID-19 Crash, Scaling and Overshoot

- Chronos-2 standardises the context by its mean and standard deviation, then applies $\sinh^{-1}$ (Ansari et al. (2025)); on 16.03.2020 the crash raised the 512-day standard deviation by 9.7% (mean absolute scale of Chronos: 3.2%, Seminar A4)
 - ▶ So the widening mixes the scale change and the reading of the crash days; only an ablation could separate them
- Not a unit error: after the crash day Chronos-2 forecast a mean $|r|$ of 12.5% for the next day, then FHS multiplies it by the 1% quantile of $r/\hat{\sigma}$ (4.7)
 - ▶ The filter extrapolates a one-day spike as a new level: VaR 4.6 times the largest loss, a high capital cost that the FZ0 loss penalises

Appendix: Berkowitz Tests and Quantile Scores, Three Markets

Market	Berkowitz p : Chronos-2	FHS	Below $q_{1\%}$ (C2)	QS: C2 / FHS	DM [p]
S&P 500	0.019	0.599	1.67%	0.510 / 0.506	+3.06 [0.002]
Bitcoin	0.595	0.306	1.17%	1.526 / 1.529	-0.90 [0.369]
BET	0.045	0.436	1.50%	0.465 / 0.460	+3.48 [<0.001]

- The Berkowitz test evaluates the linearly interpolated grid distribution function and only μ, σ , not serial independence of the PIT
- QS weights the 21 levels equally; the CRPS would weight each level by its spacing; C2: Chronos-2; tie on Bitcoin

Appendix: FZ0 Losses and MCS p -Values, Full Table

FZ0 [MCS p]	HS	GARCH-t	FHS	Chronos-2 quantiles	FHS + Chronos-2 vol.	FHS + TimesFM vol.
S&P 500	1.309 [0.14]	1.026 [0.19]	0.999 [1.00]	1.147 [0.14]	1.084 [0.14]	1.059 [0.18]
Bitcoin	2.237 [0.34]	2.192 [0.60]	2.192 [0.60]	2.181 [0.64]	2.151 [1.00]	2.160 [0.70]
BET	1.230 [0.01]	1.001 [0.34]	0.963 [1.00]	1.032 [0.24]	1.007 [0.34]	1.027 [0.20]

- Bitcoin: the hybrid FHS with Chronos-2 volatility has the lowest FZ0 loss, but the differences are not significant
- FZ0 for (VaR 2.5%, ES 2.5%); the MCS uses the $T_{\max}$ statistic and a block bootstrap

Appendix: Conformal Recalibration, Rules and Results

After 250 calibration days:					
breaches [cc p / DQ p]		FHS	Chronos-2 raw	FHS + C2 vol.	Chronos-2 + ACI
S&P 500 (24.4)	28 [0.097 / 0.014]	42 [0.002 / <0.001]	23 [0.058 / <0.001]	21 [0.307 / 0.049]	
Bitcoin (25.7)	24 [0.449 / 0.108]	31 [0.410 / 0.026]	26 [0.087 / 0.004]	24 [0.449 / 0.191]	
BET (24.2)	26 [0.709 / 0.727]	36 [0.048 / 0.184]	23 [0.776 / 0.011]	25 [0.762 / 0.032]	
FZ0, S&P 500 [DM p]	1.013	1.142 [0.009]	1.087 [0.022]	1.205 [0.001]	

- $\hat{Q}$: the $\lceil (1 - \alpha_t)(250 + 1) \rceil$ -th smallest past score, $+\infty$ if that rank exceeds 250 or $\alpha_t \leq 0$, $-\infty$ if $\alpha_t \geq 1$; with $\gamma = 0.005$: infinite VaR on 24 (S&P 500), 100 (Bitcoin), 270 (BET) days
- Guarantee with $T = 2443$, $\gamma = 0.001$: bound 0.41, uninformative for a 1% rate; without exchangeability, weighted conformal bounds the gap by the drift ([Barber et al. \(2023\)](#))
- ES comparisons are harder still: finite-sample precision limits ([Pele & Mazurencu-Marinescu-Pele \(2026\)](#)); ES is backtestable jointly with VaR ([Nolde & Ziegel \(2017\)](#)); a GPD tail ([McNeil & Frey \(2000\)](#)) is an alternative for ES

Appendix: Case Study, Replication with Course Data

- Monthly SPY total returns from 1993 (course data) in excess of the 3-month Treasury bill rate (FRED TB3MS)
 - ▶ Predictors of [Welch & Goyal \(2008\)](#) available: Treasury-bill rate (tb1), long-term yield (lty), term spread (tms), default yield spread (dfy), inflation (infl), all from FRED; stock variance (svar) from daily S&P 500 returns; the lagged excess return: 7 of the 15
 - ▶ Data limitation: the valuation ratios (dividend-price dp, dividend yield dy, earnings-price ep, payout de, book-to-market b/m), net equity expansion (ntis) and the bond returns (long-term return ltr, default return spread dfr) are not in the course data; about 30 years instead of 90
 - ▶ Method unchanged: $\gamma = 2$, $T = 12$, $P \leq 12\,000$, $\log_{10} z = -3, \dots, 3$, 1 000 draws
- Course-data results: linear kitchen sink, $z = 0^+$ ($c = 7/12$): $R^2 = -1\,540\%$, Sharpe -0.06 ; linear, $z = 10^3$: $R^2 = 0.26\%$, information ratio 0.25 ($t = 1.3$) against SPY; RFF, $c = 1\,000$: $R^2 = -6.5\%$, information ratio -0.17 ($t = -0.9$)

Appendix: The Discovery Loop, Reusable Prompts (1/2)

- **Literature:** "Find evaluations of time-series foundation models on financial returns or volatility since 2024: models, data, loss, benchmark, and whether the test period overlaps the pretraining corpus."
 - ▶ Semantic Scholar, Google Scholar, Elicit help with the search
- **Hypotheses:** "List predictions that separate memorisation from skill for a zero-shot volatility forecaster: pre vs post release, never-public series, simulated series with the same dynamics."
 - ▶ Diagnostics, not proof: a simulated log-HAR path tests transfer to that process; a pre/post gap can also reflect a regime change
- **Data and code:** "From the daily SPY realised-variance forecasts of Chronos-2 and log-HAR, compute the QLIKE differential Chronos-2 minus log-HAR, test pre vs post 3 November 2025 with HAC errors, and report the minimum detectable effect at 80% power."
 - ▶ Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot)

Appendix: The Discovery Loop, Reusable Prompts (2/2)

- **Robustness:** "Design a pre-registered test with more power: which cross-section of series, which variance proxy, which panel DM test, and how many post-release days."
- **Critique:** "Attack this claim: no memorisation, because the post-release DM test is insignificant. What is wrong with the inference?"
 - ▶ Expect: low power, absence of evidence, and different regimes: the April 2025 tariff shock lies in the pre-release sample, the post-release window is calmer
- **Report:** model versions and weight release dates, context length, mean or median, every seed and window tried

Appendix: Further Reading

- Gu, Kelly & Xiu (2020): machine learning for the cross-section of expected returns
- Kelly, Malamud & Zhou (2024): why very large models can predict returns better
- Ansari et al. (2024): Chronos, time series read as a language
- Crypto applications: liquidity forecasting Agakishiev, Härdle & Pele (2025), early warning for “zombie” crypto-assets Będowska-Sójka, Wójcik & Pele (2026), statistical classification of crypto-assets Pele et al. (2023)
- Textbooks: Franke, Härdle & Hafner (2019) and Härdle, Chen & Overbeck (2017) for the statistics of financial markets behind the benchmarks

References (1/7)

- ▣ Agakishiev, I., Härdle, W.K., Pele, D.T. (2025). [Cryptocurrency liquidity forecasting](#). In *Springer Handbooks of Computational Statistics*, 41–60. Springer.
- ▣ Aksu, T., Woo, G., Liu, J., Liu, X., Liu, C., Savarese, S., Xiong, C., Sahoo, D. (2024). [GIFT-Eval: a benchmark for general time series forecasting model evaluation](#). arXiv:2410.10393.
- ▣ Andersen, T.G., Bollerslev, T., Diebold, F.X., Labys, P. (2003). [Modeling and forecasting realized volatility](#). *Econometrica*, 71(2), 579–625.
- ▣ Ansari, A.F., Shchur, O., Küken, J., Auer, A., et al. (2025). [Chronos-2: from univariate to universal forecasting](#). arXiv:2510.15821.
- ▣ Ansari, A.F., Stella, L., Turkmen, C., Zhang, X., et al. (2024). [Chronos: learning the language of time series](#). *Transactions on Machine Learning Research*; arXiv:2403.07815.
- ▣ Barber, R.F., Candès, E.J., Ramdas, A., Tibshirani, R.J. (2023). [Conformal prediction beyond exchangeability](#). *Annals of Statistics*, 51(2), 816–845.
- ▣ Będowska-Sójka, B., Wójcik, P., Pele, D.T. (2026). [Early warning systems for cryptocurrency markets: predicting 'zombie' assets using machine learning](#). *North American Journal of Economics and Finance*, 81, 102543.
- ▣ Bengio, Y., Simard, P., Frasconi, P. (1994). [Learning long-term dependencies with gradient descent is difficult](#). *IEEE Transactions on Neural Networks*, 5(2), 157–166.
- ▣ Berkowitz, J. (2001). [Testing density forecasts, with applications to risk management](#). *Journal of Business & Economic Statistics*, 19(4), 465–474.
- ▣ Brini, A. (2026). [Forecasting realized volatility with time series foundation models: a comparison with econometric benchmarks](#). arXiv:2607.05291.

References (2/7)

- ▣ Campbell, J.Y., Thompson, S.B. (2008). Predicting excess stock returns out of sample: can anything beat the historical average? *Review of Financial Studies*, 21(4), 1509–1531.
- ▣ Christoffersen, P.F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862.
- ▣ Clark, T.E., West, K.D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1), 291–311.
- ▣ Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196.
- ▣ Das, A., Kong, W., Sen, R., Zhou, Y. (2024). A decoder-only foundation model for time-series forecasting. *Proceedings of ICML*, PMLR 235; arXiv:2310.10688.
- ▣ Diebold, F.X. (2015). Comparing predictive accuracy, twenty years later: a personal perspective on the use and abuse of Diebold–Mariano tests. *Journal of Business & Economic Statistics*, 33(1), 1.
- ▣ Diebold, F.X., Gunther, T.A., Tay, A.S. (1998). Evaluating density forecasts with applications to financial risk management. *International Economic Review*, 39(4), 863–883.
- ▣ Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Engle, R.F., Manganelli, S. (2004). CAViaR: conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4), 367–381.
- ▣ Fischer, T., Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2), 654–669.

References (3/7)

- ▣ Fissler, T., Ziegel, J.F. (2016). Higher order elicibility and Osband's principle. *Annals of Statistics*, 44(4), 1680–1707.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Garza, A., Rosillo, R., Mendoza-Smith, R., Salinas, D., Williams, A.R., Ashok, A. (2026). Impermanent: a live benchmark for temporal generalization in time series forecasting. [arXiv:2603.08707](#).
- ▣ Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- ▣ Gibbs, I., Candès, E. (2021). Adaptive conformal inference under distribution shift. *NeurIPS*; [arXiv:2106.00170](#).
- ▣ Gneiting, T. (2011). Making and evaluating point forecasts. *Journal of the American Statistical Association*, 106(494), 746–762.
- ▣ Gneiting, T., Raftery, A.E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378.
- ▣ Goel, A., Pasricha, P., Magris, M., Kannianen, J. (2025). Foundation time-series AI model for realized volatility forecasting. [arXiv:2505.11163](#).
- ▣ Gu, S., Kelly, B., Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223–2273.
- ▣ Hansen, P.R. (2005). A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4), 365–380.

References (4/7)

- ▣ Hansen, P.R., Lunde, A., Nason, J.M. (2011). [The model confidence set](#). *Econometrica*, 79(2), 453–497.
- ▣ Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (eds.) (2017). [Applied Quantitative Finance](#), 3rd ed. Springer.
- ▣ Hochreiter, S., Schmidhuber, J. (1997). [Long short-term memory](#). *Neural Computation*, 9(8), 1735–1780.
- ▣ Hornik, K., Stinchcombe, M., White, H. (1989). [Multilayer feedforward networks are universal approximators](#). *Neural Networks*, 2(5), 359–366.
- ▣ Kelly, B., Malamud, S., Zhou, K. (2024). [The virtue of complexity in return prediction](#). *Journal of Finance*, 79(1), 459–503.
- ▣ Kingma, D.P., Ba, J. (2015). [Adam: a method for stochastic optimization](#). *ICLR*; arXiv:1412.6980.
- ▣ LeCun, Y., Bengio, Y., Hinton, G. (2015). [Deep learning](#). *Nature*, 521(7553), 436–444.
- ▣ Liu, C., Aksu, T., Liu, J., Liu, X., et al. (2025). [Moirai 2.0: when less is more for time series forecasting](#). arXiv:2511.11698.
- ▣ Lo, A.W., MacKinlay, A.C. (1990). [An econometric analysis of nonsynchronous trading](#). *Journal of Econometrics*, 45(1–2), 181–211.
- ▣ Lopez-Lira, A., Tang, Y., Zhu, M. (2025). [The memorization problem: can we trust LLMs' economic forecasts?](#) arXiv:2504.14765.

References (5/7)

- ▣ McNeil, A.J., Frey, R. (2000). [Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach](#). *Journal of Empirical Finance*, 7(3–4), 271–300.
- ▣ Meyer, M., Kaltenpoth, S., Albers, H., Zalipski, K., Müller, O. (2025). [TS-Arena: a live forecast pre-registration platform](#). [arXiv:2512.20761](#).
- ▣ Nie, Y., Nguyen, N.H., Sinthong, P., Kalagnanam, J. (2023). [A time series is worth 64 words: long-term forecasting with Transformers](#). *ICLR*; [arXiv:2211.14730](#).
- ▣ Nolde, N., Ziegel, J.F. (2017). [Elicitability and backtesting: perspectives for banking regulation](#). *Annals of Applied Statistics*, 11(4), 1833–1874.
- ▣ Oreshkin, B.N., Carpov, D., Chapados, N., Bengio, Y. (2020). [N-BEATS: neural basis expansion analysis for interpretable time series forecasting](#). *ICLR*; [arXiv:1905.10437](#).
- ▣ Pan, Z., Ezzat, A.A. (2026). [Evaluating time series foundation models for electricity price forecasting: contamination risk, distributional shifts, and covariate dependence](#). [arXiv:2607.02623](#).
- ▣ Pascanu, R., Mikolov, T., Bengio, Y. (2013). [On the difficulty of training recurrent neural networks](#). *ICML*; [arXiv:1211.5063](#).
- ▣ Patton, A.J. (2011). [Volatility forecast comparison using imperfect volatility proxies](#). *Journal of Econometrics*, 160(1), 246–256.
- ▣ Patton, A.J., Ziegel, J.F., Chen, R. (2019). [Dynamic semiparametric models for expected shortfall \(and value-at-risk\)](#). *Journal of Econometrics*, 211(2), 388–413.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). [Finite-sample precision limits for expected shortfall forecast comparisons](#). *Mathematics*, 14(13), 2316.

References (6/7)

- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). *Are cryptos becoming alternative assets?* *The European Journal of Finance*, 29(10), 1064–1105.
- ▣ Rahimi, A., Recht, B. (2007). *Random features for large-scale kernel machines.* *Advances in Neural Information Processing Systems*, 20.
- ▣ Rahimikia, E., Ni, H., Wang, W. (2025). *Re(Visiting) time series foundation models in finance.* arXiv:2511.18578.
- ▣ Rasul, K., Ashok, A., Williams, A.R., Ghonia, H., et al. (2023). *Lag-Llama: towards foundation models for probabilistic time series forecasting.* arXiv:2310.08278.
- ▣ Romano, J.P., Wolf, M. (2005). *Stepwise multiple testing as formalized data snooping.* *Econometrica*, 73(4), 1237–1282.
- ▣ Rumelhart, D.E., Hinton, G.E., Williams, R.J. (1986). *Learning representations by back-propagating errors.* *Nature*, 323(6088), 533–536.
- ▣ Salinas, D., Flunkert, V., Gasthaus, J., Januschowski, T. (2020). *DeepAR: probabilistic forecasting with autoregressive recurrent networks.* *International Journal of Forecasting*, 36(3), 1181–1191.
- ▣ Tan, M., Merrill, M.A., Gupta, V., Althoff, T., Hartvigsen, T. (2024). *Are language models actually useful for time series forecasting?* *NeurIPS*; arXiv:2406.16964.
- ▣ Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., et al. (2017). *Attention is all you need.* *NeurIPS*; arXiv:1706.03762.
- ▣ Welch, I., Goyal, A. (2008). *A comprehensive look at the empirical performance of equity premium prediction.* *Review of Financial Studies*, 21(4), 1455–1508.

References (7/7)

- White, H. (2000). [A reality check for data snooping](#). *Econometrica*, 68(5), 1097–1126.
- Woo, G., Liu, C., Kumar, A., Xiong, C., Savarese, S., Sahoo, D. (2024). [Unified training of universal time series forecasting Transformers](#). *ICML*; arXiv:2402.02592.